Avaliação M1: Banco de Dados

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Contextualização

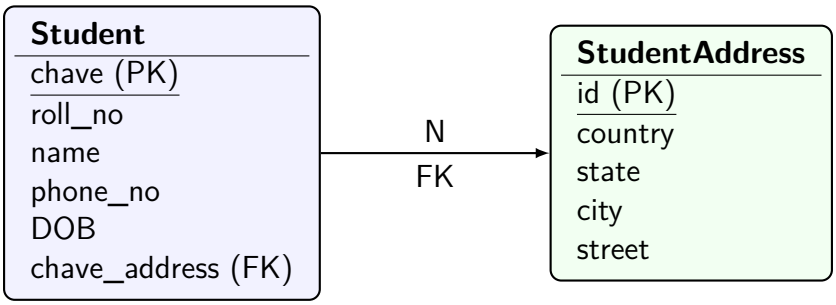
Este relatório apresenta os diagramas e scripts SQL da Avaliação M1 da disciplina de Banco de Dados (Prof. Dr. Rafael Queiroz Gonçalves).

1 Questão 1: Entidade `Student`

Enunciado: Escrever o código SQL para definir a entidade estudante a partir do diagrama ER e elaborar o diagrama lógico correspondente. Atributos `VARCHAR`, com exceção de `Roll_No` (`INT`).

Diagrama Lógico Relacional

O modelo aplica a 1ª Forma Normal (1FN). Os atributos compostos de endereço foram separados na entidade independente `StudentAddress`. A relação utiliza uma chave estrangeira (FK) na tabela `Student`.



Código SQL

```sql

-- Tables
CREATE DATABASE `2026-2_banco-de-dados`;
USE `2026-2_banco-de-dados`;

CREATE TABLE StudentAddress (
    id INT AUTO_INCREMENT PRIMARY KEY,
    country VARCHAR(50) NOT NULL,
```

```
    state VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    street VARCHAR(50) NOT NULL
);

CREATE TABLE Student (
    chave INT AUTO_INCREMENT PRIMARY KEY,
    roll_no INT NOT NULL,
    name VARCHAR(50) NOT NULL,
    phone_no VARCHAR(50),
    DOB VARCHAR(50) NOT NULL,
    chave_address INT,
    FOREIGN KEY (chave_address) REFERENCES StudentAddress(id)
);

-- Transaction

START TRANSACTION;

INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Hauptstra e , 120');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (101, 'Carlos Eduardo', '48999999999', '2000-05-15', LAST_INSERT_ID()
    );


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Bahnhofstra e , 45');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (102, 'Ana Beatriz', '11988888888', '2001-08-22', LAST_INSERT_ID());


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Neue Stra e , 78');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (103, 'Lucas Henrique', '21977777777', '1999-12-10', LAST_INSERT_ID()
    );


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Brazil', 'Santa Catarina', 'Florian polis', 'Rua das Palmeiras,
    120');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (104, 'Mariana Silva', '41966666666', '2002-03-18', LAST_INSERT_ID())
    ;


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Karlstra e , 230');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (105, 'Gabriel Souza', '31955555555', '2000-11-27', LAST_INSERT_ID())
    ;


INSERT INTO StudentAddress(country, state, city, street)
```

```
VALUES ('Brazil', 'S o Paulo', 'S o Paulo', 'Avenida Paulista, 1500');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (106, 'Juliana Oliveira', '51944444444', '2001-06-09', LAST_INSERT_ID
    ());


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'M nsterplatz, 15');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (107, 'Rafael Martins', '71933333333', '1998-09-14', LAST_INSERT_ID()
    );


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Brazil', 'Paran ', 'Curitiba', 'Rua XV de Novembro, 780');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (108, 'Beatriz Costa', '81922222222', '2002-01-30', LAST_INSERT_ID())
    ;


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Olgastra e, 310');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (109, 'Pedro Augusto', '85911111111', '2000-07-25', LAST_INSERT_ID())
    ;


INSERT INTO StudentAddress(country, state, city, street)
VALUES ('Germany', 'Baden-W rttemberg', 'Ulm', 'Einsteinstra e, 90');

INSERT INTO Student(roll_no, name, phone_no, DOB, chave_address)
VALUES (110, 'Larissa Almeida', '48900000000', '2001-10-05', LAST_INSERT_ID
    ());

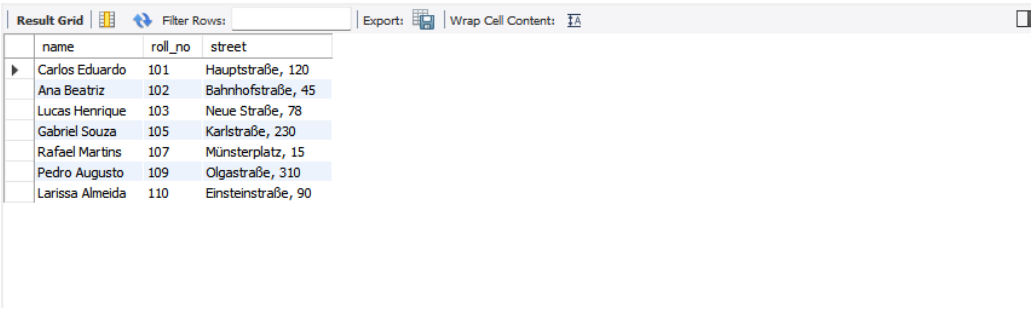
COMMIT;
```

2 Questão 2: Projeção de Estudantes em Ulm

Enunciado: Retornar estudantes que moram em "Ulm", projetando `Name`, `Roll_No` e `Street`.

Código SQL

```
SELECT name,
       roll_no,
       (SELECT street
        FROM StudentAddress
        WHERE StudentAddress.id = Student.chave_address) AS street
FROM Student
WHERE chave_address IN (
    SELECT id
    FROM StudentAddress
    WHERE city = 'Ulm'
);
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	roll_no	street
Carlos Eduardo	101	Hauptstraße, 120
Ana Beatriz	102	Bahnhofstraße, 45
Lucas Henrique	103	Neue Straße, 78
Gabriel Souza	105	Karlstraße, 230
Rafael Martins	107	Münsterplatz, 15
Pedro Augusto	109	Olgastraße, 310
Larissa Almeida	110	Einsteinstraße, 90

Figura 1: Query retornada

3 Questão 3: Modelo ER Musical

Enunciado: Código SQL para o diagrama ER Musical (`Artist`, `Album`, `Track`). `played` é do tipo `DATETIME`, chaves secundárias são `INT`, e colunas de texto são `VARCHAR`.

Código SQL

```sql
USE `2026-2_banco-de-dados`;

-- Tables

CREATE TABLE Artist (
    artist_id INT PRIMARY KEY,
    artist_name VARCHAR(50) NOT NULL
);

CREATE TABLE Album (
    album_id INT PRIMARY KEY,
    album_name VARCHAR(50) NOT NULL,
    artist_id INT NOT NULL,
    FOREIGN KEY (artist_id) REFERENCES Artist(artist_id)
);

CREATE TABLE Track (
    track_id INT PRIMARY KEY,
    track_name VARCHAR(50) NOT NULL,
    time INT NOT NULL,
    album_id INT NOT NULL,
    FOREIGN KEY (album_id) REFERENCES Album(album_id)
);

CREATE TABLE Played (
    played DATETIME NOT NULL,
    track_id INT NOT NULL,
    PRIMARY KEY (track_id, played),
    FOREIGN KEY (track_id) REFERENCES Track(track_id)
);

-- Transaction

START TRANSACTION;

INSERT INTO Artist(artist_id, artist_name) VALUES
(1, 'Linkin Park'),
(2, 'The Weeknd'),
```

```
(3, 'Imagine Dragons'),
(4, 'YOASOBI'),
(5, 'Bruno Mars');


INSERT INTO Album(album_id, album_name, artist_id) VALUES
(1, 'Meteora', 1),
(2, 'Hybrid Theory', 1),
(3, 'After Hours', 2),
(4, 'Starboy', 2),
(5, 'Evolve', 3),
(6, 'Night Visions', 3),
(7, 'THE BOOK', 4),
(8, '24K Magic', 5);


INSERT INTO Track(track_id, track_name, time, album_id) VALUES
(1, 'Numb', 185, 1),
(2, 'Breaking the Habit', 196, 1),
(3, 'In the End', 216, 2),
(4, 'Papercut', 184, 2),
(5, 'Blinding Lights', 200, 3),
(6, 'Save Your Tears', 215, 3),
(7, 'Starboy', 230, 4),
(8, 'The Hills', 242, 4),
(9, 'Believer', 204, 5),
(10, 'Thunder', 187, 5),
(11, 'Radioactive', 186, 6),
(12, 'Demons', 177, 6),
(13, 'Yoru ni Kakeru', 261, 7),
(14, 'Racing into the Night', 248, 7),
(15, '24K Magic', 226, 8),
(16, 'Thats What I Like', 206, 8);


INSERT INTO Played(played, track_id) VALUES
('2026-09-01 08:30:00', 1),
('2026-09-01 10:15:00', 1),
('2026-09-02 14:20:00', 1),

('2026-09-01 09:00:00', 2),
('2026-09-03 16:45:00', 2),

('2026-09-01 11:30:00', 3),
('2026-09-02 18:00:00', 3),
('2026-09-04 20:15:00', 3),

('2026-09-02 08:45:00', 4),

('2026-09-01 13:00:00', 5),
('2026-09-02 15:30:00', 5),
('2026-09-05 21:00:00', 5),

('2026-09-03 19:20:00', 6),
('2026-09-05 22:10:00', 6),

('2026-09-01 17:00:00', 7),
('2026-09-04 12:40:00', 7),

('2026-09-02 20:00:00', 8),
```

```
('2026-09-01 07:30:00', 9),
('2026-09-03 18:30:00', 9),
('2026-09-05 23:00:00', 9),

('2026-09-02 09:45:00', 10),

('2026-09-03 10:30:00', 11),

('2026-09-04 14:00:00', 12),

('2026-09-01 22:30:00', 13),
('2026-09-03 21:15:00', 13),
('2026-09-05 20:45:00', 13),

('2026-09-02 23:00:00', 14),

('2026-09-04 17:30:00', 15),
('2026-09-05 19:00:00', 16);

COMMIT;
```

4 Questão 4: Restrição por Tempo de Duração

Enunciado: Projetar o nome e tempo das faixas com `time > 180`.

Código SQL

```
SELECT track_name, time
FROM Track
WHERE time > 180;
```

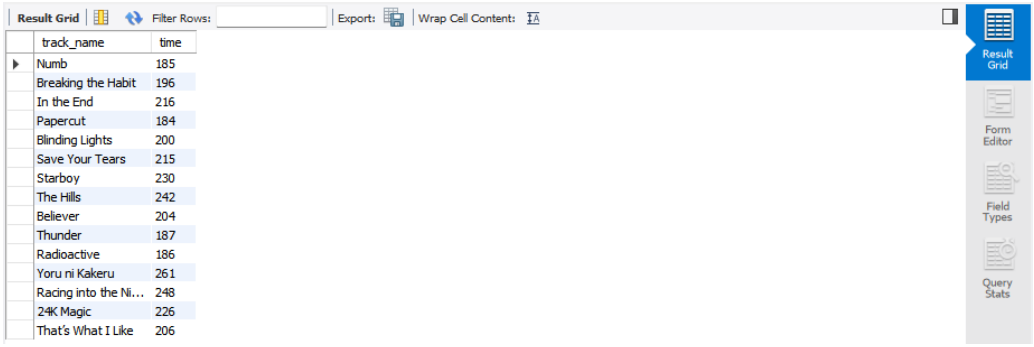


Figura 2: Query retornada

5 Questão 5: Histórico de Reproduções

Enunciado: Dado o id de uma faixa, projetar o histórico de reproduções.

Código SQL

```
SELECT played
FROM Played
WHERE track_id = 1;
```

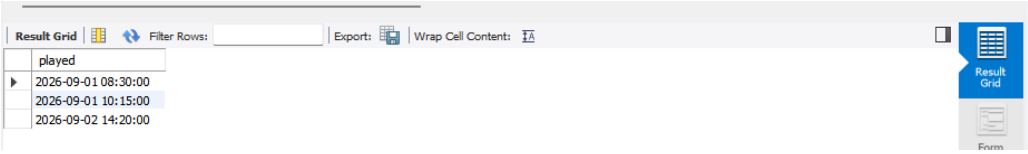


Figura 3: Query retornada